RAJALAKSHMI ENGINEERING COLLEGE

Department of Information Technology

II Year – III Semester





MINI PROJECT PRESENTATION

COLLEGE CANTEEN ORDERING SYSTEM

Byte And Bite

Course: Object-Oriented Programming and Database Management Systems

TEAM MEMBERS

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Introduction

Our project, ByteAnd Bite – College Canteen Ordering System, is a Java-based application developed using Object-Oriented Programming (OOP) concepts and MySQL database connectivity. It allows students to view the food menu, place orders for multiple food items, calculate the total bill, and update the available food quantity in the database. The main aim of this project is to make the college canteen ordering process simple, organized, and efficient.

Problem Statement



Managing food orders, calculating bills, and maintaining food stock manually in a college canteen can be time-consuming and may lead to errors. During busy hours, students may experience delays in placing orders and receiving their bills. Manual record-keeping also makes it difficult to maintain accurate food availability and order details.

ByteAnd Bite provides a simple, menu-driven canteen ordering system developed using Java and MySQL. It allows students to order multiple food items in a single transaction, calculates the total bill automatically, and updates food stock in the database. By combining Java OOP concepts with database connectivity, the system reduces manual work, improves order accuracy, and makes canteen management more efficient.



Our Innovative Solutions

Technologies Used

- Programming Language: Java
- Database: MySQL
- Connectivity: JDBC (Java Database Connectivity)
- Concepts: Object-Oriented Programming (OOP)
- Development Environment: VS code terminal
- Database Tool: MySQL Workbench





Software Requirements

Operating System: Windows

Programming Language: Java (JDK)

Database: MySQL Server

Database Tool: MySQL Workbench

Connectivity: MySQL JDBC Connector

Development Environment: Visual Studio Code (VS Code)

Execution: VS Code Terminal





System Design – ByteAnd Bite

Student / User
Selects food items and quantities

ByteAndBite.java
Main class · Starts the application



MenuManager.java
Displays menu · Processes orders · Calculates bill

FoodItem.java
Stores food item details, price, and quantity



JDBC Connectivity
Connects Java application to MySQL

MySQL Database via JDBC
Stores food and order details and updates stock

DBMS Schema – ByteAnd Bite

Database: byteandbite

Table: menu

itemid —primary key, INT, AUTO_INCREMENT
itemname — VARCHAR(100), NOT NULL
category — VARCHAR(50), NOT NULL
price — DECIMAL(10,2)
availability — INT, NOT NULL

Result Grid | Filter Rows: | Edit:

itemid	itemname	category	price	availability
1	Masala Dosa	Breakfast	70.00	50
2	Idli	Breakfast	40.00	42
3	Parotta	Breakfast	15.00	50
4	Paneer Dosa	Breakfast	100.00	50
5	Chapati with Paneer Curry	Breakfast	120.00	47
6	Chicken Biryani	Lunch	150.00	150
7	Veg Fried Rice	Lunch	100.00	150
8	Chicken Rice	Lunch	130.00	150
9	Tomato Rice	Lunch	60.00	150
10	Curd Rice	Lunch	50.00	150
11	Samosa	Starters	20.00	200
12	Momos	Starters	80.00	200
13	Sandwich	Starters	60.00	200
14	Pani Puri	Starters	40.00	200
15	Shawarma	Starters	120.00	200
16	Pizza	Fast Food	150.00	200
17	Burger	Fast Food	100.00	200
18	White Sauce Pasta	Fast Food	120.00	200
19	Hakka Noodles	Fast Food	100.00	200
20	Lays	Chips	NULL	200
21	Bingo	Chips	NULL	200
22	Doritos	Chips	NULL	200
23	French Fries	Fried Items	70.00	195
24	Fresh Lime Juice	Juices a…	30.00	200
25	Rose Milk	Juices a…	40.00	200

13	Sandwich	Starters	60.00	200
14	Pani Puri	Starters	40.00	200
15	Shawarma	Starters	120.00	200
16	Pizza	Fast Food	150.00	200
17	Burger	Fast Food	100.00	200
18	White Sauce Pasta	Fast Food	120.00	200
19	Hakka Noodles	Fast Food	100.00	200
20	Lays	Chips	NULL	200
21	Bingo	Chips	NULL	200
22	Doritos	Chips	NULL	200
23	French Fries	Fried Items	70.00	195
24	Fresh Lime Juice	Juices a…	30.00	200
25	Rose Milk	Juices a…	40.00	200
26	Mango Milkshake	Juices a…	80.00	200
27	Strawberry Smoothie	Juices a…	100.00	200
28	Mango Lassi	Juices a…	70.00	200
29	Gulab Jamun	Chocolat…	30.00	157
30	Brownie	Chocolat…	80.00	200
31	Ice Cream	Chocolat…	50.00	200
32	Chocolate Cake	Chocolat…	80.00	200
33	Falooda	Chocolat…	100.00	196
NULL	NULL	NULL	NULL	NULL

```
111 • SELECT itemid, itemname, availability
112   FROM menu
113   WHERE itemid = 5;
```

Result Grid | Filter Rows: | Edit:

itemid	itemname	availability
5	Chapati with Paneer Curry	47
NULL	NULL	NULL

Description: The menu table stores food item names, categories, prices, and available quantities. The database supports category-wise menu display, availability checking, and menu record counting.

OOP Implementation – ByteAnd Bite



Class and Object

FoodItem.java defines food items, and objects represent individual items.

Encapsulation

Food item data can be grouped within the FoodItem class.

Abstraction

MenuManager.java organizes menu and ordering operations behind methods.

Modularity

ByteAndBite.java, MenuManager.java, and FoodItem.java separate application responsibilities.

Role of Each Java File

- FoodItem.java: Represents food item details such as name, price, and availability.
- MenuManager.java: Manages menu display, order processing, and bill calculation.
- ByteAndBite.java: Contains the main method to start the application.

Output Screenshots – ByteAnd Bite

menu card of ByteAndBite





```
PS C:\Users\thami\OneDrive\Desktop\ByteAndBite\src> java -cp ".;../lib/mysql-connector-j-26.7.0.jar"
ByteAndBite

===== BYTE AND BITE FOOD MENU =====
ID    Food Item                 Category                  Price     Available
-------------------------------------------------------------------------
1     Masala Dosa               Breakfast                 70.00     50
2     Idli                      Breakfast                 40.00     50
3     Parotta                   Breakfast                 15.00     50
4     Paneer Dosa               Breakfast                 100.00    50
5     Chapati with Paneer Curry Breakfast                 120.00    50
6     Chicken Biryani           Lunch                     150.00    150
7     Veg Fried Rice            Lunch                     100.00    150
8     Chicken Rice              Lunch                     130.00    150
9     Tomato Rice               Lunch                     60.00     150
10    Curd Rice                 Lunch                     50.00     150
11    Samosa                    Starters                  20.00     200
12    Momos                     Starters                  80.00     200
13    Sandwich                  Starters                  60.00     200
14    Pani Puri                 Starters                  40.00     200
15    Shawarma                  Starters                  120.00    200
16    Pizza                     Fast Food                 150.00    200
17    Burger                    Fast Food                 100.00    200
18    White Sauce Pasta         Fast Food                 120.00    200
19    Hakka Noodles             Fast Food                 100.00    200
20    Lays                      Chips                     0.00      200
21    Bingo                     Chips                     0.00      200
22    Doritos                   Chips                     0.00      200
23    French Fries              Fried Items               70.00     200
24    Fresh Lime Juice          Juices and Drinks         30.00     200
25    Rose Milk                 Juices and Drinks         40.00     200
26    Mango Milkshake           Juices and Drinks         80.00     200
27    Strawberry Smoothie       Juices and Drinks         100.00    200
28    Mango Lassi               Juices and Drinks         70.00     200
29    Gulab Jamun               Chocolates and Desserts   30.00     200
30    Brownie                   Chocolates and Desserts   80.00     200
31    Ice Cream                 Chocolates and Desserts   50.00     200
32    Chocolate Cake            Chocolates and Desserts   80.00     200
33    Falooda                   Chocolates and Desserts   100.00    200
```

single order placing, order confirmation ,billing and updation of stock.



```
1. Place Order
2. Exit
Enter your choice: 1

Enter Food Item ID: 2
Enter Quantity: 4

====== ORDER CONFIRMED ======
Food Item : Idli
Quantity  : 4
Total     : Rs. 160.00
Stock updated successfully!
```



updated menu

```
----- UPDATED MENU -----

========== BYTE AND BITE MENU ==========
ID    Food Item                 Category                  Price       Available
-------------------------------------------------------------------------------
5     Chapati with Paneer Curry Breakfast                 120.00      50
2     Idli                      Breakfast                 40.00       46
1     Masala Dosa               Breakfast                 70.00       50
4     Paneer Dosa               Breakfast                 100.00      50
3     Parotta                   Breakfast                 15.00       50
21    Bingo                     Chips                     Not set     200
22    Doritos                   Chips                     Not set     200
20    Lays                      Chips                     Not set     200
30    Brownie                   Chocolates and Desserts   80.00       200
32    Chocolate Cake            Chocolates and Desserts   80.00       200
33    Falooda                   Chocolates and Desserts   100.00      200
29    Gulab Jamun               Chocolates and Desserts   30.00       200
31    Ice Cream                 Chocolates and Desserts   50.00       200
17    Burger                    Fast Food                 100.00      200
19    Hakka Noodles             Fast Food                 100.00      200
16    Pizza                     Fast Food                 150.00      200
18    White Sauce Pasta         Fast Food                 120.00      200
23    French Fries              Fried Items               70.00       200
24    Fresh Lime Juice          Juices and Drinks         30.00       200
28    Mango Lassi               Juices and Drinks         70.00       200
26    Mango Milkshake           Juices and Drinks         80.00       200
25    Rose Milk                 Juices and Drinks         40.00       200
27    Strawberry Smoothie       Juices and Drinks         100.00      200
6     Chicken Biryani           Lunch                     150.00      150
8     Chicken Rice              Lunch                     130.00      150
10    Curd Rice                 Lunch                     50.00       150
9     Tomato Rice               Lunch                     60.00       150
7     Veg Fried Rice            Lunch                     100.00      150
12    Momos                     Starters                  80.00       200
14    Pani Puri                 Starters                  40.00       200
11    Samosa                    Starters                  20.00       200
13    Sandwich                  Starters                  60.00       200
15    Shawarma                  Starters                  120.00      200
PS C:\Users\thami\OneDrive\Desktop\ByteAndBite\src>
```

My sql connector is connected.

The output for the jdbc connection



```
PS C:\Users\thami\OneDrive\Desktop\ByteAndBite> java -cp "out;lib/*" DBConnection
================================
          BYTE AND BITE
================================
MySQL Connector Connected Successfully!
```



Multiple order placing,billing,updated menu availability.

```
PS C:\Users\thami\OneDrive\Desktop\ByteAndBite\src> java

1. Place Order
2. Exit
Enter your choice: 1

Enter Food Item ID (0 to finish): 33
Enter Quantity: 4
Added: Falooda
Item total: Rs. 400.00

Enter Food Item ID (0 to finish): 23
Enter Quantity: 5
Added: French Fries
Item total: Rs. 350.00

Enter Food Item ID (0 to finish): 29
Enter Quantity: 43
Added: Gulab Jamun
Item total: Rs. 1290.00

Enter Food Item ID (0 to finish): 5
Enter Quantity: 3
Added: Chapati with Paneer Curry
Item total: Rs. 360.00

Enter Food Item ID (0 to finish): 2
Enter Quantity: 4
Added: Idli
Item total: Rs. 160.00

Enter Food Item ID (0 to finish): 0

===== ORDER CONFIRMED =====
GRAND TOTAL: Rs. 2560.00
```









updated menu availability

```
----- UPDATED MENU -----

========== BYTE AND BITE MENU ==========
```

ID	Food Item	Category	Price	Available
5	Chapati with Paneer Curry	Breakfast	120.00	47
2	Idli	Breakfast	40.00	42
1	Masala Dosa	Breakfast	70.00	50
4	Paneer Dosa	Breakfast	100.00	50
3	Parotta	Breakfast	15.00	50
21	Bingo	Chips	Not set	200
22	Doritos	Chips	Not set	200
20	Lays	Chips	Not set	200
30	Brownie	Chocolates and Desserts	80.00	200
32	Chocolate Cake	Chocolates and Desserts	80.00	200
33	Falooda	Chocolates and Desserts	100.00	196
29	Gulab Jamun	Chocolates and Desserts	30.00	157
31	Ice Cream	Chocolates and Desserts	50.00	200
17	Burger	Fast Food	100.00	200
19	Hakka Noodles	Fast Food	100.00	200
16	Pizza	Fast Food	150.00	200
18	White Sauce Pasta	Fast Food	120.00	200
23	French Fries	Fried Items	70.00	195
24	Fresh Lime Juice	Juices and Drinks	30.00	200
28	Mango Lassi	Juices and Drinks	70.00	200
26	Mango Milkshake	Juices and Drinks	80.00	200
25	Rose Milk	Juices and Drinks	40.00	200
27	Strawberry Smoothie	Juices and Drinks	100.00	200
6	Chicken Biryani	Lunch	150.00	150
8	Chicken Rice	Lunch	130.00	150
10	Curd Rice	Lunch	50.00	150
9	Tomato Rice	Lunch	60.00	150
7	Veg Fried Rice	Lunch	100.00	150
12	Momos	Starters	80.00	200
14	Pani Puri	Starters	40.00	200
11	Samosa	Starters	20.00	200
13	Sandwich	Starters	60.00	200
15	Shawarma	Starters	120.00	200

THANK YOU!!